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H-SCOOP

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BITS & PIECES

H-SCOOP ISSUE

I get an occasional call about where the issue number is on this newsletter. I usually explain it every few issues for the benefit of new subscribers. It is important to know to keep track of your subscriptions and to use the general yearly index. First off, the issue # is located on the upper right corner of the newsletter, such as ISS#2303020282. The first two digits are all you are mainly concerned with, 23 is issue # 23 which is February. The second two, 03 is the volume number, the next two, 02 is the issue this year, the next two is the month (02= February), and the last two is the year.

H-100 BITS

More information flows in on the new super computer. It is said that it will have four different operating systems. Modified HDOS and CP/M which are the 8 bit operating systems. The 16 bit operating systems are CP/M 86 and an unnamed one that will allow the H-100 to run the IBM Personal Computer software. The H-100 is rumored to be able to handle both 8 and 16 bit. It is also to have more memory than the H89, and in spite of all this, the price should be right around what the H89 is selling for. I was also told that this project is the only of higher priority than the double density controller for the H8.

HEATH H8 Z80 CPU

In the January issue I printed some information on patching the UltiMeth SY.DVD to work with the Heath Z80 CPU. Brian Hansen writes in "I am using the Z80 card with the HUG version of the new SY.DVD (this is Dean Gibson's first release--ed) and have had no problems with it. I am using two standard 100K Siemens drives and no other features other than selectable step speeds and head load times. I think that you will find on the Z80 card (as mine was), the jumper from E4 to E5 is not on the board. E4 to E5 is the ROM Size Select. With E4 to E5 IN the ROM area is 0 to 4K and the second ROM Socket U19 Rom#2 is disabled. If E4 to E5 is OUT then the ROM Select area is 0 to 8K and socket U19 ROM #2 is active. I believe that an access to the 4K to 8K area with E4 to E5 out will try to read or write to an empty ROM Socket U19."

Brian goes on to comment on the side select with the Z80 card. "It can be set up as the Preferred Method #1 as on the 0 Org board doing the following.

1. Make sure that no NONHEATH boards use BUSS Pin #18!
2. On the back of the H-17 Controller board, Jump S103 (ribbon connector) Pin #32 to S102 (card edge) Pin #18.
3. On the Z80 card Jump E46 (side select) to E38 (Buss Pin 18)".

I prefer the jumper method rather than the buss #18 method since this signal is only used for one card. Why drag it along the entire buss--ed.

Brian further writes "I have also found that the HA8-6 Z80 Manual is lacking in information. I had purchased the manual back in August 1981 and found it had many errors. I wrote Barry Watzman and told him of the errors. His reply confirmed all the errors and he had passed the information on to the Manual Department. When my board finally arrived in late October NONE of the errors were CORRECTED and NO ADDENDUMS were in the manuals. The Schematic that I received still has ERRORS on it. Some of the I.C.'s are drawn without the inversion symbols on their output pins. I found U3, U20c-d, U22e, U11a and there might be others."

That's just the kind of support from Heath we need! Take your time to point out errors, and they SIT ON IT! I take this space to make this point because I receive this same type of response from my readers from time to time. Heath also wants you to send in warranty cards so they can keep you updated on changes and corrections made. As far as I know, this doesn't happen. I purchased one of the earlier parallel boards. The manual had many errors in it and was very lacking in information. Some readers sent in corrections to me that they also sent in to Heath, and I never received word of these changes, which would have made life much easier had I received them. I purchased a new manual much later and saw some changes made in the manual, but I was never notified by Heath! I guess they are getting to busy to support the ones that support them!!!

With Brian's comment of the manual "lacking in information", I also have a comment. It seems that Heath's manuals, which once were known for their total information and clarity no longer exist. At one time, with a Heath manual you could easily understand how the equipment worked and troubleshoot it if you had the proper knowledge and equipment. The manuals I received with my H19 and H89 left much to be desired. There's no way I could do much with those manuals. Even though these systems contain some "Heathbuilt boards" they should still explain them and have some trouble shooting procedures so those more advanced could take care of their own systems. If the Heathkits keep going in the direction they have been going, they will have to change the name from Heathkit to Heathbuilt! I realize many boards are complex and require calibration, but they may be going overboard. Enough said.

Now that I have that off my back, I return to the Z80 with information from Larry Wier. Larry writes that he had problems with the Tandon drives and the Z80 CPU with INIT and SYSGEN. From Larry, "I got INIT to work by using the original HEATH 'INIT' with UltiMeth SY.DVD. I don't know why it worked but it did. By the way the info in the Jan issue is in error. Using the Ver. 1 SY.DVD, the last paragraph under "TANDONS WITH HEATH H8 Z80" and at the end of line 4 it should read as follows:
---track 0 and sector 0 starting with address 5B should---
The words "0 starting with address" are missing." The best thing is to obtain the latest version from myself or Dean of UltiMeth as that has this change plus other enhancements incorporated in it. It is available as a free update to those owners of Version 2 by exchanging the disk, or for $20 for those with the first release of SY.DVD. Please note that source code is no longer supplied in versions above 1.0.

BSR-X-10 AND H8-2 PARALLEL

Larry Wier also tells me that the October issue of **73 MAGAZINE** (a HAM magazine) on page 116 has an article on interfacing the H8 to the BSR-X-10 via the H8-2 parallel board. I guess Larry is the author. He has available the software and a copy of the interface building procedures if anyone is interested. It is written for MBASIC/HDOS or CASSETTE EX BH BASIC, please specify which one you want when ordering. A SASE will get details, and $10 (specify disk or tape) will get you a copy of the software. It's free in the magazine.

BITS

I talked to Bob Ellerton of REMark and he told me that **HDOS 3.0** will support the H8 as well as the H89. The only reason it is not being advertised for sale for the H8 is because the H8 presently does not support double density drives or hard disks from Heath, which was the primary reason for the HDOS 3.0 release.

SUPERCALC should be released in a version which will be PIPable to different drives etc, after February. I guess Sorcim's trick to try to prevent Pirating has made many unhappy, as they cannot even use it on their own systems if the disk configuration is different than the SUPERCALC distribution disk. Those who purchased the old version, don't despair. Check TECHNICAL FORUM for a way to patch your version to make it PIPable.

NEW STUFF

MAIL LIST PROGRAM

Dave Powers of GENERIC SOFTWARE/ POB 1154/ TROY, MI 48099/ (313) 879-1115 has available under HDOS or CP/M a mail list program called **MAIL-IT**. It will run on an H89 or H8 with H19 and disk. MAIL-IT is a comprehensive system of programs to maintain name, address, and phone number information. It utilizes disk for all user data and not RAM, which means you are limited by disk system. It will accommodate users with high density minis, 8" floppys, or even hard disks. Any number of records can be specified for the database within the limits of disk storage capacity.

MAIL-IT uses random access disk files and three separate retrieval techniques, (1) random access by record number, (2) random access using an index file based on the Last Name, and (3) sequential search on any field using a user supplied substring. It is menu driven to provide a turnkey operation. It provides sorting by Name and Zip code, and allows record selection based on Last Name, Zip code, and/or user-defined flags. Sorting is limited by memory size, however a small efficient sort like M-KEY SORT (past H-SCOOP offer) would work nicely with this program.

Each record in the mail list database contains: Last Name, First Name, Address, City, State, 9 digit Zip code, Home phone, Work phone, and user defined Flags. MAIL-IT requires at least 160K disk storage and an MBASIC interpreter. It is available for $34.95 which includes 5" disk, 20 page doc, and shipping and handling. Documentation is available separately for inspection for $5 which can be applied to later purchase. Specify HDOS or CP/M. Write or call for more info.

SOFTWARE TOOLWORKS DISK FORMATS

The Software Toolworks now has their software products available on 5" soft sector disks. All their 21 software products can be ordered on soft sector disks for the Z90 and other systems using the Z-89-37 soft sector controller. Each product is available in the same disk formats as for hard sector: CP/M, HDOS and/or Dual HDOS-CP/M, depending on the product. For a copy of their catalog, or for more info, write or call The Software Toolworks/ 14478 Glorietta Dr/ Sherman Oaks, CA 91423/ (213) 986-4885 weekdays between 11 and 5 P.S.T.

Also from Software Toolworks is an HDOS printer Spooler which supports all disk types. It is compatible with all disk devices and drivers running under HDOS 2.0, including the UltiMeth drivers. Other added features from previous releases included expanded SET options to configure the spooler to drive Heath, Diablo, Epson, and just about any other serial driven printer. It is available for $29.95 plus $2 postage and handling. Specify hard or soft sector when ordering.

TERMINAL .ACM FILE

William Hall of MLM Associates/ 5621 Maple Heights Ct/ Pittsburgh, PA 15232 has available a new ACM file, HEATH19.ACM, for assembly programmers. Since the various escape sequences used by Heath or ANSI have mnemonics which are listed in the documentation, the ACM file was created. It gives the values of the Heath mnemonics for such things as "Save cursor position", "Enter reverse video mode", "Go to 25th line", "Write message beginning an column n", "Exit reverse video", "Return cursor". If programming in assembly, this will make life much easier when utilizing the terminal escape sequences. It is available for $9.95, or $6.95 if you supply disk. The package consists of the disk with .ACM file, two page summary sheet mounted in plastic with a list of all terminal functions, each with its corresponding escape sequence and Heath mnemonic. In addition, the terminal hardware switches are shown, the sequences transmitted by the shifted and alternate keypad, special key sequences, and a complete ASCII chart in decimal, octal and hex. Thrown in is the comparable file for MACRO 80 users.

CP/M DATA BASE MANAGEMENT

Zenith Data Systems has introduced CONDOR Series 20 rDBMS, a relational data base management software package. The Condor program will be of interest to business people who frequently change the way in which they are analyzing info, or who need the ability to easily add new items to their info base. This system organizes lists or files of information together in a relationship defined by th user.

It uses clear English commands, making the establishment of data bases and retrieval of data easier. Since it is written in assembly language, it is very fast. Because it is implemented under CP/M, it is compatible with Magic Wand and WordStar word processing programs. Built in 'HELP' aids are also available. It is available for $895 on 8" and DD 5.25" disks for use with H37. Contact a Zenith Data System dealer to purchase, or for more information.

UNIT CONVERSION SOFTWARE

Mako Data Products/ 1441 #B N. Red Gum/ Anaheim, CA 92806/ (714) 632-8583 has some exciting news. First is a CP/M software disk which is a universal convert anything to anything calculator. **UNIT CONVERSION MASTER** (UCM) is one of the most comprehensive sets of conversion tables ever assembled. Convert over 550 units of measurement within 20 different categories including Length, Area, Volume, Velocity, and Time--more than 18,000 possible conversions. It is not just a list of factors, it will automatically convert your input to ALL other units within the selected category.

UCM is easy to use and has graphic screen entry, and full help facilities. It's something like an electronic spreadsheet (SuperCalc type). You enter the item you want, and instantly all other related output items are updated. Very impressive. This is one of the better and more useful software works I have seen. Definitely a MUST for teachers, students, experimenters, whoever. The only feature I would like to see on something like this, is also the conversion factor formulas, which would be a great learning aid for children and adults learning metric conversions etc.

Although you'd expect to pay a lot for something like this, it is available for only $19.95 post paid USA, foreign orders add $2.00. It requires CP/M 2.2 (it's a .COM file) with 64K memory, and is distributed on a 5" SD disk. Since graphics is used, an H8/H19 or H89 is required.

H8 DD DISK CONTROLLERS

Mako Data Products also is releasing an H-37 (H89) compatible double density board for the H8 computer. The MH8-37 double density soft sector disk controller should be available late February, giving up to 200K free per disk using the SS SD ST 5" drives, and up to 782 sectors free with the 96TPI (Tandon DS DD DT) drives. The board is a 100 percent functional equivalent to the Z-89-37 upgrade

Heath offers for the H89, and runs the EXACT same operating system (CP/M 2.2.03). Mako will be working on an HDOS upgrade to allow HDOS to also be able to use this board. Presently, it is only for CP/M users.

Skip of Mako Data Products tells me this board will work on H8's with the 8080 CPU board, and uses the Western Digital 1797 Disk Controller chip. Buss space for this board is free since it replaces the 0-org at the extreme rear of the H8. The board has the latched 8 bit user status port which may be optionally configured as a 0-org mod for those with the original 8080 CPU. It also has the necessary side select on it, according to Skip, so the 0-org board is no longer needed when this board is installed.

They guarantee hardware compatibility with all available 2 and 4 Mhz Z80 CPU boards. Because their H8 and H89 PSG sound boards have proven so reliable Mako is extending all warranties to a full year from date of delivery. This warranty will also apply to the MH8-37 disk controller.

Skip also tells me this board will handle either four 5" drives, or four 8" drives. It has the plug connections for each on the board. Price? First 100 sold will go for $330 each, and after that, $380. Add $5 to each order for shipping and handling, (foreign orders, add $8). Send orders to Mako Data Products, 1441-B N. Red Gum, Anaheim, CA 92806. Be sure to include your CPU board type. For phone inquiries, call (714) 632-8583.

As a bit of comparison, I felt I should repeat some info I printed on the Livingston PC17 double density 5" controller in the January issue (page 2). The PC17-8 (for H8) and PC17-89 (for H89) are replacement boards for the H-17 single density controller for the H8 and H89 which will allow the use of both single and double density from the same card. This means you can still get the single density, and no longer need your present single density controller. While Mako's board will use soft sectoring to be H37 compatible, Livingston's board will use the same 10 hole hard sector disks you presently use, making it unnecessary to change disks, although it will not be H37 compatible. Since it is 100% compatible with the single density board, all old disks will still be able to be BOOTed with this card, and all existing software will function exactly as before. The PC17 will support up to 4 5" drives in any combination of 40 tracks, to 80 track DT. Maximum storage is 200K for the standard SS ST drives, and 800K for the Tandon type 96 TPI DS DD DT drives. To make life easier, it will automatically detect single and double density disk densities, allowing both single and double to be used in the same drive.

To be released shortly, the expected prices are $300 for the PC17-8 version (H8) and $275 for the PC17-89 version. These prices will include either CP/M or HDOS driver (specify which you want). HDOS will run under HDOS 2.0, and CP/M under CP/M 2.2.3. Add $30 if you want the other driver. Both the Mako board and Livingston board are sold completely assembled. The Livingston board price is lower. One reason for this is that you must take some chips out of the present H17 board and place them in the PC17 board. Oh yes, as mentioned previously, this board requires a Z80 CPU board. For more info on this one, contact Ray Livingston, of Livingston Logic Labs/ POB 5334/ Pasadena, CA 91107, or phone (213) 792-4763.

AND MORE FROM RAY

Don't forget to contact Ray if you are interested in the double density board for the H8 that will handle up to four 5" drives and 4 8" drives. As mentioned in January, this will be about a $450 board. It will require Z80 CPU, and will work with both HDOS and CP/M. He will go into production if he receives enough feedback to let him know it will be a worthwhile endeavor. If you are interested, write, or forever hold your tongue. I'd hate to see this not get off the ground because you didn't take the time to write or call Ray!

Ray also wants me to pass along this information. He has worked out with Dean Gibson of Ultimeth to be able to supply the famous UltiMeth MTRXXX Monitor ROM for use with his 8" controller. For a description of what these ROM's do, refer to November 1981 H8SCOOP page 2, and December issue, page 3. Basically, you get increased HDOS processing speed, 2 or 4 MHz sensing and processing (when used with appropriate drivers), better RAM testing, extended TICCNT clock to 4 bytes, and debugging of assembly programs under HDOS. I have installed one in place of my Magnolia monitor ROM. I can now boot off any HDOS device. I could not run REACH with my Magnolia ROM, but with the UltiMeth ROM I can. If you want to order one for your H89 in OCTal or HEX, for the H89, contact Dean of UltiMeth Corp/ 24025 Fernlake Dr/ Harbor City, CA 90710. The price is $50. If you want one for the 8" controller from Ray Livingston, order either the MTROCT or MTRHEX from Ray, the price still $50.

MONITOR AND CONTROL SYSTEMS

ICS/ POB 14571/ Minneapolis, MN 55414 has available some RS-232 driven controller cards. The SRIO connects to a host computer by RS-232 serial port and can operate up to 256 BSR system X-10 remote control modules. Single quantity price is $275 without clock, and $325 with clock.

They also have talkers, A/D converters and more. The R001 to R400 analog, status and control intelligent remote modules provide remote sensing and remote control with feedback to central control (host) computers when used with the SRIO. Each unit can be configured with a mixture of up to 4 channels. Write them for more info.

BUSINESS SOFTWARE

From Integrated Business Systems/ POB 438/ Powers Lake, WI 53159/ (414) 279-6706 comes a variety of business software for use on the Heath/Zenith Z89. They have software available in CP/M and CBASIC2, FORTRAN, and COBAL for A/R, Order Entry, G/L, A/P, Payroll, Inventory, Job Order Entry, Estimating, Loan Amortization, Medical Billing, and others. They are not yet, but may be configured on the H37 format DD DS disks. If you are interested in finding out what they are up to, drop them a line, or call.

NEWLINE SOFTWARE/ POB 402/ Littleton, MA 01460 is distributing the S & M Systems Inc. business software. Write them for compete details. They have Accounts Payable, Accounts Receivable, General Ledger with Cash Journal, Inventory control, and Payroll for $125 each. General requirements are 64K system with 0 org CP/M, MBASIC 5.1, minimum 2 5.25" drives, or 2 8" drives, and 132 column printer.

Also available from NEWLINE is a new version of **VIDEO SCRIBE**. It is a text editor for use with H8/H19 or H89's with 32K minimum memory and 1 drive, under HDOS 1.6 or higher. Several bugs in the old releases, including the one that most bothered me, uneven spacing when justifying, have been corrected. A new algorithm for WRAPping and PADding now evenly distribute space when justifying text. A new CENTER command has been added to center text. A new feature allows switching between the unshifted keypad commands and a new NUMERIC entry command. Another new feature is the selection of compressing spaces to TABs, which can be turned on or off. There is even an OOPS key to undo your mistakes.

VIDEO SCRIBE is the most powerful text editor in its price range. There are more than 50 commands for entering, editing, moving, searching, and filing your text. The price is $39.95 and can be ordered from NEWLINE or your local Heath store. To examine the documentation before buying, send $1. A DEMO version with documentation is also available on 5.25" disks for $6. All registered owners of VIDEO SCRIBE will receive a discount coupon and ordering instructions in the mail. Unregistered owners are not eligible.

Two MORTGAGE SCHEDULERs are available. HAT SOFTWARE/ POB 27402/ HONOLULU, HI 96827 has a package written in FORTRAN. It is an accurate and very fast mortgage or loan program. Price is $12.45 which includes shipping and handling. It requires 132 column printer. Since it is written under HDOS FORTRAN, it will run stand alone as .ABS code. The program MORT.ABS will do a 30 year mortgage schedule in 15 seconds, compared to 2 minutes for a BASIC version. To run it you need H8 or H89 with 1 drive, 1.6 or 2.0 HDOS, and 132 column printer. You can print to terminal or printer. You enter amount borrowed, interest, and total number of years on mortgage. It calculates what you owe, INTerest and PRIncipal paid on the loan.

A similar program, under CP/M as a .COM file is LOANPAY.COM for $17 which includes shipping and handling. This program is an improvement over most loan amortization programs in that is rejects inappropriate inputs, and provides an option for tracking the loan if one cares to pay more than the required monthly payment. It uses double precision mathematics to insure the output is correct to the penny. It calculates monthly payment, interest and principal paid each month, and to date. It also provides yearly totals of interest and principal payments so it can be used to help prepare tax returns, and is therefore, tax deductible. LOANPAY requires 24K, 80 character printer, one drive and CP/M operating system. It is supplied on single density 5.25" disk, just like the above. This is available from Bruce Bennett of Mountain Software Co/ 299 Smith Hill Rd/ Utica, NY 13502.

HAT software sent me a sample mortgage table, and Mountain sent me the LOANPAY program. I ran it with the values that HAT used for their table, and got the same results. I was very impressed with the program Mountain sent me. If you want to make a few extra bucks with your system, or just want to help figure your loan payments on that dream house, or new computer you want to buy, these look like good programs. How to make a few bucks? This would be a good service to provide for Real Estate people, if they do not have a computer (I know, I have already been contacted by one for this service--before I had the program!!).

CHECKING ACCOUNT PROGRAM

James Meyers/ 13A Riggs Pkwy/ Las Vegas, NV 98115 has available CHECKFIL.BAS, a menu driven program written in MBASIC 4.82 under HDOS 1.6 or 2.0. The purpose of CHECKFIL is to supervise a home checking account in an effective and fast manner. James says that although there are scores of programs that do the same thing, what good is a program if you spend 30 minutes fiddling with it when the job only takes you 10 minutes with a pencil and paper (see my editorial)? With CHECKFIL you only need enter a CR to input you check number because it features auto numbering of checks and adjustments. Balancing can be done in 3 minutes, and if it doesn't balance, CHECKFIL will help you find the error with its Search capability. It also backs up files automatically, and reads the backup if the original is damaged.

You can edit, sort, merge, catalog, delete, scan the file and more. It also contains a .CON(figure) file that change the program parameters from H19 format to H9, from dual drive to single, or from 48K to 56K. It comes with over 100 sectors of documentation. If you desire information, send a SASE to Jim. Price is $30, shipping included in USA. Foreign orders add $3. It is distributed on 5.25" hard sector disk, and requires 48K memory.

UPDATE ON H89 SOUND BOARD

Ray Albrektson/ 619 W. Dover/ San Bernardino, CA 92407/ (714) 882-8072 is shipping the first Sound Clock Calendar board for the H89. He wants to let all my readers know that they come with complete instructions for installing them on either side of the H89. When installed on the left side, a few signals must be "stolen" from some right-side board, but hooking it up is straight forward. He also offers a no quibbling money back guarantee: If buyers give it their best shot and it won't work, he'll buy it back with no questions asked.

For those that missed my previous writeup, this board enables the user to cheaply interface a bus oriented digital clock/ calendar to the H89 by means of two 8 bit parallel ports available on the popular AY3-8910 PSG. The total outlay should be about $30 parts plus price of the bare board. Full access to the time, date, year is available from BASIC and other languages. System software under development can update HDOS date on boot, etc. The PSG provides the usual 3 tone/ noise channels, envelope generator, etc. Program listings are also provided for setting and reading the clock as well as a modified CK.DVD for an automatic on-screen clock. Bare board price is $22 which includes complete instructions and all program listings. Complete instructions and all listings (to wirewrap your own) available for $5. All programs on 5" HDOS disk $5 with purchase of $22 kit, otherwise $15. You get all these in printed form free with either of the above, this is just for the sake of not having to type them in!

Another nifty device Ray is offering is an AUTO BOOTER, which is a small circuit board which installs under the keyboard of the H/Z89. It waits for power to be turned on, and a few seconds later, it electronically pushes the "B" and the "CR" keys to begin the boot. He developed this so he could turn on the H89 with a BSR X-10 timer, have the machine boot itself, and run automatically. The possibilities are even more intriguing since BSR recently introduced a phone activated controller. With a clock calendar board and suitable prologue, the H89 can check the time and date and know what program to run after waking up.

Although there is a small risk of destroying the disk files by having a disk in the machine on power up and down, Ray has had no problem with this. He always keeps a backup disk just in case. This board sells for $22 in kit form, postpaid, and can be assembled in an hour. Shipping date is February 10th.

CASSETTE SOFTWARE

Those waiting for good cassette software, here it is. L.I.R.S.-8 is a Library Information Retrieval System which should configure on any H8 with greater than 8K memory. This program allows entry of articles and books with the assignment of a unique FILE NUMBER to each. To allow subsequent reference, searches by AUTHOR/ EDITOR, TITLE, FILE NUMBER, FREE STRING SEARCH or by subject HEADERS. It was written on an H8 with 32K and an H9 terminal, H8-5 cassette interface, and TED-8. LIRS-8 is a stand alone operating system which uses no H8 ROM subroutines. The source code listing may be entered and assembled on any 8080, Z80, 8088 or 8086 CPU with an assembler. Source code is included on side 2 of the tape, and is about 1300 lines of code. It should also work on a cassette H88 if it can be booted. The program assumes the ROM in the lower 8K, so starts at 040.100A. This could easily be changed by changing the ORG statement and re-assembling. The top 128 bytes are also left alone since the H8 uses about the top 80. There is no memory overflow routine, but Total File Summary will warn anyone if you are close to top memory.

The program does not yet support a printer as the author does not have one, but it could easily be added. Future updates will provide printer support, and perhaps disk support.

The tape also will include some front panel routines for the H8. They include flashing LED's and clicks for warning before deleting files of numbers greater than those in actual use. The cost of this tape is $15 which includes tape, manual, and mailing. This is a lot of program and extras for this price. To order, or for more information, contact Larry Carlson/ 10834 Dixon Dr. So/ Seattle, WA 98178.

BIBLIOGRAPHY UPDATE

Microcomputing, January 1982 had two Heath related articles. Page 150 by Charles Cohn **HEATH HIDDEN TIME-SAVER** tells how to extract the date from the HDOS system to use in programs. This is nothing new, but it was in the magazine so I thought I'd mention it.

On page 72 by Robert Bradley is **Tapping Into the Brain.** An assembly language program is listed, using the H8 with an included interface circuit to gather information on brain function. The actual brain wave pickup circuit is not given.

WHO'S WHO

This column is not getting the usage I thought it would when I formulated the newsletter format. This column is to help those keep in touch with others doing similar work or unique work with their computers to help to some extent, the re-inventing of the wheel. If you are using your computer for a specific or unique thing, for instance, A/D conversion, household control etc., please drop me a line so I can print it and keep others in touch.

Have you modified your system to do a specific function, or to make it more efficient? Let us know!

For those of you who are interested in computer graphics on the H8, **Wayne Frick/ 1433 Dennison Av/ Staunton, VA 24401** has devised the software and hardware to interface an H8 to an oscilloscope with X and Y inputs to create graphics on it. It was patterned from an article in BYTE magazine, November 1976 titled "Build the Beer Budget Graphics Interface. It uses three 74100 latches, and two 1408L8 eight bit D/A converters for 256 X 256 resolution. It can be easily set up on the H8-7 breadboard card. It's an excellent experimenters project. As many as 1500 dots may be displayed before flickering becomes too apparent.

I must apologize for the delay on this mention. I received this info about a year ago, and was going to publish it. Then space began to get very scarce, and I felt the article may be beneficial to some H8ers, but not a general interest article, so I decided not to print it. I did want to let you know it has been done, and very well indeed. If you are interested in this project, contact Wayne. He has prepared about a 20 page article on it, any may make copies available for a small charge.

There is now a **PHILADELPHIA HUG.** It meets at 7:00 on the second Wednesday of each month at the Heathkit Electronic Center at Roosevelt Blvd, with Hank Beechhold its president. Phone (215) 288-0180.

REQUESTS

John of John's Schwinn Cyclery/ 1656 E. State St/ Sharon, PA 16146 has just purchased SUPERCALC and is very impressed with it. He wants to know if anyone or any group wishes to exchange information on this program, if so, contact him. Since SUPERCALC is very powerful, and quite amazing, it wouldn't be a bad idea to exchange ideas through this newsletter. I feel sooner or later any serious computer individual or business will have it. If you have a good idea or unique application, send it in, and I'll publish it.

Rod Madsen/ 9502 Mary Circle/ Villa Pk, CA 92667 wants to know if anyone knows how to read a **SUPERCALC** file (under CP/M) into a page editor under HDOS.

Barry Gottlieb/ 4804 Patterson Av/ Richmond, VA 23226/ (804) 359-0001 has purchased an H90 with the double density card and is having trouble getting his Tandon 100-4 up and running with it. If anyone has gone this route and has any information, please get in contact with Barry.

Mark Jackson/ 16 Ironside St/ St. Lucia 4067/ QLD./ AUSTRALIA is interested in contacting H8 user groups in the United States with an interest in participating in group purchase schemes such as Lucidata Pascal and H8 hardware. He says the Australian agents are friendly, but also slow and ineffective. He also tells me that they are about to start their own independent users group.

OFFERS

The next complete listing of offers available will be in the April issue of H-SCOOP as it is only printed quarterly to conserve space. For those interested the price of the Magnolia H89 DD controller board with CP/M has been dropped to $499. The price of 100-4 Tandons is $425 for one, and $415 for the second. Anyone who purchased drives previously can take advantage of the quantity prices.

I am totally re-working the software offers. I will be eliminating some from the list, and combining some, as to keep my inventory down, and give you more for your money. Hopefully the April listing will reflect this.

This month I have the 2/82 offer, which is a take-off of the 10/81 offer. Anyone who puchased the 10/81 400K dup, dump package can receive this by sending your disk back plus $2. The cost of this will be $10. The 10/81 release of the dup and dump for the 400K drives only worked with the release 1 of SY.DVD from UltiMeth. They both have been modified to let them work with the new releases, and add more features. The DUMP will let you access all 160 tracks of the Tandon drives. The DUP will allow you to DUP between two similar drives, from 40 Track SS through 80 Track DS.

The author, Terry Smedley tells me the Dup and Dump programs are capable of all the the functions of DUPs and DUMPs which are selling for substantially more. Both programs will recognize any type of valid media in any drive (40's in 80Tk drives etc). The DUP program will also display the exact track and sector of any bytes found different between two disks upon verification. DUP also displays the volume number and label of any destination disk it finds which is not blank. The DUP also has a LED display, for those with the H8, so the user can see what track they're on during copy and verification.

All necessary ACM's are included for re-assembling the programs, along with the source code for the programs. Also included is the Serial Selectric device driver which operates with the H8-4 card and has SETable options as to CPU speed, baudrate, port, and echo or noecho to console. SCREEN.BAS is an enhanced and modified program as found originally in REMark issue 12. The idea behind this is to make screen formatting easier in BASIC. Refer to issue 12 of REMark for details. This version makes use of special function keys and much more. It was submitted by William Gunckel.

In addition to these, which were on the 10/81 disk, some files are included with H8 LED routines. FPLED(.ASM, .ABS, .LST) is a demo of software control of the H8 front panel from an assembly program. It makes use of a complete 'micro-library' of subroutines found in H8LEDS.ACM which allows the user to easily control the LED's from an assembly language program. There are two pages of documentation found at the beginning of the .ACM file. Routines are included which allow the user to select no LED activity for maximum throughput; user specified LED messages; or software controller PAM-8 displays. As an example, MVI A,BC and CALL LEDREG is sufficient to cause the LED's to display the current contents of the BC register pair. Also included is a complete library of MBASIC subroutines which allow software control of the H8 LED's from MBASIC. Also is a program, DISKERR.BAS which will display the global error counters from HDOS in an MBASIC program. This will allow you to get disk statistics from MBASIC.

A MUST FOR H8ERS, OR 400K-ERS EXCELLENT!

I have been told that my disk offers section was too vague, so this is my first attempt to try to more thoroughly explain what the offer contains and what the programs are and do. I hope it helps. I will try to follow this format in the future. That's all for now.

PROGRAMMING GOODIES

For those who have not figured this out yet, to get a hard copy listing of a program from BH BASIC, while still in the BASIC, do the following:
REPLACE "AT:" (or LP:)
In MBASIC SAVE "AT:",A (or LP:)

For doing long division on a computer in BASIC, try the following:
Say A=97 and B=5
Q=INT(A/B)
R=A-B*Q
Q(uotent) is the # times B goes into A
R(emainder) is what's left over from A when you divide B

When working with strings, remember BASIC leaves 1 character space in front of a number (positive) for the sign, even though not printed.

To save memory when writing a BASIC program, place multiple commands on one line where possible. Omit all spaces, and delete all REMarks. Keep in mind that REMarks and spaces use memory.

To cause a BASIC program to run faster; using LET A=5 uses more memory, but will usually run faster than saying A=5.

To enter programs easier in MBASIC, using ? instead of PRINT and ' instead of REMark will make life easier.

This next goodie was taken from HUGNJ newsletter, October 1981 issue. It was written by Francis Stepper/ 54 Commerce St/ Garfield, NJ 07026.

FRAME IT!

Sometimes, it is desirable to box in, or frame, portions of the screen to set it off from the rest, making it auxiliary text or making that section prominent. Following is a software routine for making a box to surround a portion of the screen. All that is required is to enter the values for the dimensions of the box (height and length) and the place to position the upper left corner. The program is written in the form of a demonstration, and has not been compacted to just a few lines of code, so that you may better understand it.

The entries for row and column are corrected automatically by adding 31 in line 180. The character used is found in line 170; you can select others, but I used reverse video in line 160 so beware. This was done so that the box lines were of even thickness. Also, keep in mind that whenever you don't follow the print statements with a semicolon, the text to the right is erased, so you may wish to enter text first, otherwise part of the box will be erased.

```
10 REM - DRAW A BOX
20 E$=CHR$(27)
30 E1$=E$+"E"
40 F$=E$+"F"
50 G$=E$+"G"
60 P$=E$+"p"
70 Q$=E$+"q"
80 Y$=E$+"Y"
90 PRINT E1$
100 REM - START BOX AT A SPECIFIC LOCATION
110 REM - ASK FOR DIMENSIONS
120 REM
130 INPUT "Where do you want box to start
    (give row and column)";R,C
140 INPUT "What are the dimensions
    (horiontal, vertical)";H,V
150 PRINT P$;
160 PRINT F$;
170 Z$="p"
180 R=R+31:C=C+31:R1=R:C1=C
190 FOR Z=1 TO 2
200 PRINT Y$;CHR$(R);CHR$(C);
210 FOR X=1 TO H
220 PRINT Z$;
230 NEXT X
240 R=R+V
250 PRINT Q$;
260 NEXT Z
270 PRINT P$;
280 FOR Z=1 TO 2
290 R=R1
300 FOR X=1 TO V-1
310 PRINT Y$;CHR$(R+1);CHR$(C);" ";
320 R=R+1
330 NEXT X
340 C=C+H-1:R=R1
350 NEXT Z
360 PRINT Q$;G$
```

DUMP ASCII TO DIABLO 630

D. C. Shoemaker sent in a short programs to allow dumping an ASCII file from HDOS to a Diablo 630. This routine is in MBASIC 4.82. D. C. says it is nothing fancy, but it might save someone from reinventing this particular wheel. It also can save the need to reset the LP.DVD and reboot.

```
100 'THIS PROGRAM WAS WRITTEN IN MBASIC 4.82
110 'UNDER 2.0 HDOS USING LPH44.DVD RENAMED
    AS LP.DVD
120 '         BY D. C. SHOEMAKER
140 PRINT CHR$(27);"E":'ERASE H19/H89 SCREEN
160 CLEAR 5000:'RESERVE STRING SPACE
170 PRINT"  THIS PROGRAM PRINTS ANY TEXT
     FILE TO THE DIABLO, ALLOWING "
180 PRINT"THE PRINTER TO PAUSE AT THE
     BOTTOM OF THE PAGE FOR A FRESH SHEET"
190 PRINT"OF PAPER TO BE INSERTED. AS
     WRITTEN, THE PROGRAM GENERATES 55 LINES"
200 PRINT"PER PAGE.":PRINT:LINE INPUT"WHAT
     FILE DO YOU WANT?    ";F$
210 '
220 'OPEN FILES FOR I/O
230 OPEN "I",#1,F$
240 OPEN "O",#2,"LP:"
280 FOR I=1 TO 55:'LINES/PAGE
290 IF EOF(1) THEN 340
300 LINEINPUT#1,A$:PRINT#2,A$:NEXT I:
     CLOSE#2:'READ AND PRINT A LINE
310 PRINT "PUT ANOTHER SHEET IN THE
     PRINTER AND PRESS ";
320 LINEINPUT"RETURN WHEN READY: ";R$
330 GOTO 240:'GO BACK FOR NEXT SET OF LINES
340 PRINT:PRINT"  FINISHED WITH ";F$;"."
360 END
```

I edited this a bit to make shorten it by removing some REMark lines, but it's basically the same program.

I also have all sorts of programs in Pascal to set printers, but since I am doing a column for REMark magazine, they will be printed there.

TECHNICAL FORUM

DISK DRIVE LUBE

I have received feedback on lubricating disk drives. For the screw type stepper head system place a very small amount of a teflon base lubricant on the screw mechanism and move the head back and forth to spread the teflon and work it in. When you are finished, be sure to return head to track 0 position (this will cause contact to be made with the track 0 microswitch.

Some drives like the Tandons have a band stepper mechanism. Some of these drives are also quite noisy on stepping. To quite them, remove the top circuit board and apply a small amount of teflon lubricant on the metal guide rods that the head mechanism travel on. Work the head back and forth and return it to track 0 position when finished. This type of lubricant is available at most gun smith type shops under the name of TRI-FLO or other various names.

H8 WITH H8-7 AND Z80

This item from Bill Hall of MLM Associates/ 5621 Maple Heights Ct/ Pittsburgh, PA 15232. If you have the H8-7 breadboard card and like to interface with the H8, you may be interested in this article, especially if switching to a Z80 CPU since I/O to ports is done differently.

"Without adding any components you can wire the board to give you a parallel port addressed from port 0 to 255 and these can be accessed either by a memory or an input/ output read. Furthermore, one of the data lines can be latched on output, and readers wanting some details can write me for more information. For example, I use the H-8-7 to drive a small relay which can be used to turn a variety of things on and off.

"But the H-8-7 has a serious limitation if you have a Z-80 CPU. However, it can be corrected. When the 8080 does an input/ output instruction, it puts the port address on the upper address lines A8-A15 and the lower lines A0-A7. The Z-80 also puts the output port address on lines A0-A7. But the content of the accumulator is put on the upper address lines. And on an indirect port read/ write the contents of register B goes on the upper lines. This offers no problems if the lower address lines are buffered and decoded. However, on the H-8-7 the UPPER address lines are buffered and decoded. This means that if you have your board wired to decode port 010Q(8) say, then whenever you have this number in the accumulator and do a port read, whatever the port being read, you will also read port 8. It can damage disks and generally mess up a program.

"There are two ways to correct it. The board also offers lines A0-A7, but they are unbuffered and not decoded. You can, however, construct your own decoder and buffers on one of the breadboards. The second solution is to cut the lines on the board at the 25 pin input socket from A8-A15 and then solder wires from the lower address lines, A0 to A8, A1 to A9, Etc. Thus, for example, a wire is run from the solder point on A7 to pin 1 of IC 104, and the line from pin 1 to the bus connector is cut. The other lines are wired similarly. I have done this and the board works fine with my DG Z-80 CPU. I ran the wires on the back of the board to reduce clutter. Thus, even Z-80 users can take advantage of this inexpensive way to build a parallel port.

HOW TO PIP SUPERCALC

I have received quite a few duplicate notices downloaded from MNET on how to PIP SuperCalc to different disks, by Pat Swayne. Doc Campbell sent one in and says you will need SDUMP from HUG (885-1213). This is public domain information and is for giving away, not for sale. To access this info, from SIG/Access: TYP SCPIP.DOC. Howard Honig also sent me a copy, and so did Bob Ellerton. Many thanks for the feedback and concern.

The SuperCalc (TM) program currently being sold by Heath cannot be copied using PIP. While this does not rule out making copies of the entire disk using DUP, it means that you can't copy from one type of media to another (eg. from a 5-inch disk to an 8-inch disk). The problem is that SC.COM (SuperCalc) is actually two files with the same name, and when you try to PIP them, you only get the first one. If the directory is patched so that the second file has a different name, then both files can be copied to a new disk, and the new disk can be patched to restore SC.COM to its original format. I will not go into detail about how the CP/M directory works, etc. but I will present the patches necessary to copy SC.COM.

To make the patches, you will need SDUMP.COM from HUG disk no. 885-1213. Read the SDUMP instructions (in SDUMP.ASM) carefully if you are not familiar with it. If you have two drives, duplicate the distribution disk, and make the patches on the duplicate.

If you have only one drive, you will have to patch your distribution disk, so use caution. SYSGEN the disk (as explained in the SuperCalc Manual), and copy SDUMP.COM to it. After you have prepared your disk, use SDUMP to make the following patch on Track 3 Sector 2 if it is a 5-inch disk, or Track 2 Sector 2 if it is an 8-inch disk. (Howard Honig says if you have soft sector DD disks, the location is also Track 2 and Sector 2).

Change Address 23 from 20 to 31.
Change Address 2C from 06 to 00.
Change Address 43 from 20 to 31.
Change Address 4C from 07 to 01.

Exit SDUMP using control-C and do a DIRectory on the disk. In addition to the three .CAL files, you should see SC.COM and SC1.COM. Prepare the disk that you will copy using SuperCalc to. It should be a newly formatted disk with no files on it that has not been SYSGENed. Use PIP to copy the SC files as in the following example:
PIP B:=A:SC?.COM
If you have a single drive system, you should now SYSGEN the new disk and copy SDUMP.COM to it. Then use SDUMP to make the following patch to Track 3 Sector 1 if the disk is 5-inch, or Track 2 Sector 1 if it is 8-inch. This patch is valid for all disk sizes except 8-inch SD DS disks.

Change Address 43 from 31 to 20
Change Address 4C from 00 to 06
Change Address 63 from 31 to 20
Change Address 6C from 01 to 07

If your new disk is 8-inch SD DS make the following patch to Track 2 Sector 1.
Change Address 23 from 31 to 20
Change Address 2C from 01 to 07

You can now SYSGEN the new disk (if you have already don it), and copy the .CAL files or any other files you wish to it. If you made the first patch to your distribution disk and wish to restore it to original condition, re-do the first patch, but reverse the numbers in the from to columns. GOOD LUCK.

H8 CRASH FIXES

I have reported previously on cures for crashing H8 computers. This was mostly because of the inferior motherboard Heath provided with the H8, and the earlier versions with the tin connectors. The best fix was to substitute the motherboard with the Trionyx gold pin ground plane mother board. My system has been operating for about a year now with this board, and has not gone down. Jim Benesch send me some additional information, which I felt you may find useful.

Jim says he installed the Trionyx motherboard, but his system still crashed. "Next I went through each card and installed condensers to filter out noise on the power sources for critical chips (timers, oneshots etc.). Values used were .1, .05, and .01 (by varying the values, the filtering effects covers a wider frequency of the noise spectrum). Again, this helped but still occasional and unpredictable crashes.

"Then I replaced the masked ROM from Heath, ordered to replace the slower EPROM sent with the Extended Configuration Option. Things really improved but still crashed.

"Finally I began replacing the edge connectors with gold (also available from Trionyx). When I replaced the 8080 CPU card edge connector, my system crashed 10 seconds after power-up. This time it stayed down. I began looking for bad chips with a can of cold spray. When I froze a 74LS240 on the CPU card, everything came up. To verify that the chip was bad, I put a hot soldering iron next to it - the system crashed. I replaced the chip and have not had a crash since.

"I think the gold connectors made a better connection resulting in a heavier load from the bus affecting the 74LS240 bus driver. The increased load changed an intermittent problem to a permanent one. Looking back, I think I might have been able to find this problem faster using hair dryer heat on each chip. This is dangerous, however, since excessive heat can damage good parts. Needless to say, intermittents are far harder to find than permanent component failures. I'd be happy to discuss this further with anyone having similar problems."

If you want to talk to Jim, he can be reached at POB 1811/ Minden, NV 89423, or phone (702) 266-3686.

CLASSIFIEDS

FOR TRADE--Assembled H19 with cables. Never used. Need Modem, H8 boards, Disk software (must be original disk). Will consider cash, Ham gear, and H8 accessories. Phone (214) 572-4712 after 6PM. Ceasar Dominguez.

WANTED--Used Z89 computers for under $1500. Contact Allen Bernson/ POB 68/ Boston, MA 02167.

PEEKING & POKING

OVERBOARD

I'll try to make this editorial brief, but I feel there's a very important point to make, and that point is going overboard and not using common sense. I've already heard of individuals loading up a tape (or booting a disk), bringing their system up, loading BASIC, and running a program. So what? Well, this was for reading a database to obtain a single telephone number! Exaggeration? No! I've also seen this done for retrieving a recipe. Let's use our sense. Things like looking up a recipe or looking up a telephone number are much faster doing it by hand than using a computer, especially if it were not even "on line"!

What about things like appointment calendars? Fine, if you have the system up and running constantly, with that program always in memory. But if not, a calendar using a pen is faster and more efficient. Let's face it. What business executive running a business program will stop the program, load in a calendar program, and search through it to see his appointments for the day? Likewise, who will go through the same thing to enter an appointment? I can do the same thing in 10 seconds on a standard calendar that the program takes minutes to do!

A computer is a great thing, but let's don't forget that some things can be done better and more efficiently without one.

Here's my actual point. It concerns the MNET (or any other for that matter) bulletin board. Some things are better handled with a personal letter. Others are better handled by a telephone call to a specific individual. If you want some info from a person, it will cost you more to get on line with CompuServe and leave a message than a long distance phone call would have cost, and you'll get an immediate answer. I'm concerned because I feel the bulletin boards are being used too much. This concerns me because that's all the less interaction that will appear in newsletters like this one (thus why NEW STUFF) is beginning to take over this newsletter. Many individuals don't have MODEMs or can't afford the long distance charge and on line computer charge, preventing them from accessing a bulletin board. What about them?

So valuable information gets passed along on the BB and the newsletters suffer. Say you have a problem and leave it on the BB for a response. Chances are, dozens of individuals around the world (Yes, I said around the world!) are having the same problem. They may get a newsletter, but will never learn what was on the MNET BB. Which brings me to another problem.

The MNET is getting so crowded that the entire message data base changes within a weeks time. That means if you are on the MNET you have to access it more than once a week to not loose messages or not miss important information. If you can't make it that week, too bad, it's all gone. Nothing left for reference. With a newsletter, it's there always, for constant referral and posterity! There is lot of information being lost that way that may never get out. If that's the way this Tandon drive matter was handled, most of you who purchased drives from me, and learned about the 400K situation would still be sadly running your 100K drives. There is a great value in the newsletter, and let's not sell it short, ever. It's your exchange and your current information media.

I have just about dropped off the Bulletin Boards. I can't even get a free line to the system till 2 or 3 in the morning (which is when I write all me editorials). If I'm lucky enough to get on, on weekends, I find I have already missed messages and other pertinent information, which I constantly scan for, for this newsletter. It's getting so bad it doesn't even pay to get on anymore. Scanning the messages takes about 15 minutes, and you find they are mostly items which could have been handled better by phone! You have to be on over a half hour, if you're fast, to get the info. For long distance callers like myself, that's over $10 to retrieve hardly anything useful.

I hope you can understand my point. I love BB's and enjoy being on them but I feel we are all going overboard, and unless this straightens up, we'll be headed for trouble (like expenses and wasted time). Have any comments, let me know.

Henry

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